Data Structures and Algorithms · 21CSC201J

Unit 3 · Session 5 · SLO 2 — Solved Worksheet

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Unit â€“ 3 : Session â€“ 5 : SLO â€“ 2 | Evaluating a Postfix Expression

I. Algorithm to Evaluate a Postfix Expression

• Step 1: Scan the postfix expression from left to right.
• Step 2: If the symbol is an OPERAND, PUSH its value onto the stack.
• Step 3: If the symbol is an OPERATOR, POP two operands (first pop = operand2, second pop = operand1),
• compute operand1 OPERATOR operand2, and PUSH the result back.
• Step 4: Repeat until the expression ends.
• Step 5: The single value remaining on the stack is the result.
• Step 6: EXIT

II. Worked Example: 9 3 4 * 8 + 4 / -

• 9 -> push. Stack: [9]
• 3 -> push. Stack: [9, 3]
• 4 -> push. Stack: [9, 3, 4]
• * -> pop 4, 3; 3*4 = 12; push. Stack: [9, 12]
• 8 -> push. Stack: [9, 12, 8]
• + -> pop 8, 12; 12+8 = 20; push. Stack: [9, 20]
• 4 -> push. Stack: [9, 20, 4]
• / -> pop 4, 20; 20/4 = 5; push. Stack: [9, 5]
• - -> pop 5, 9; 9-5 = 4; push. Stack: [4]
• Result: 4